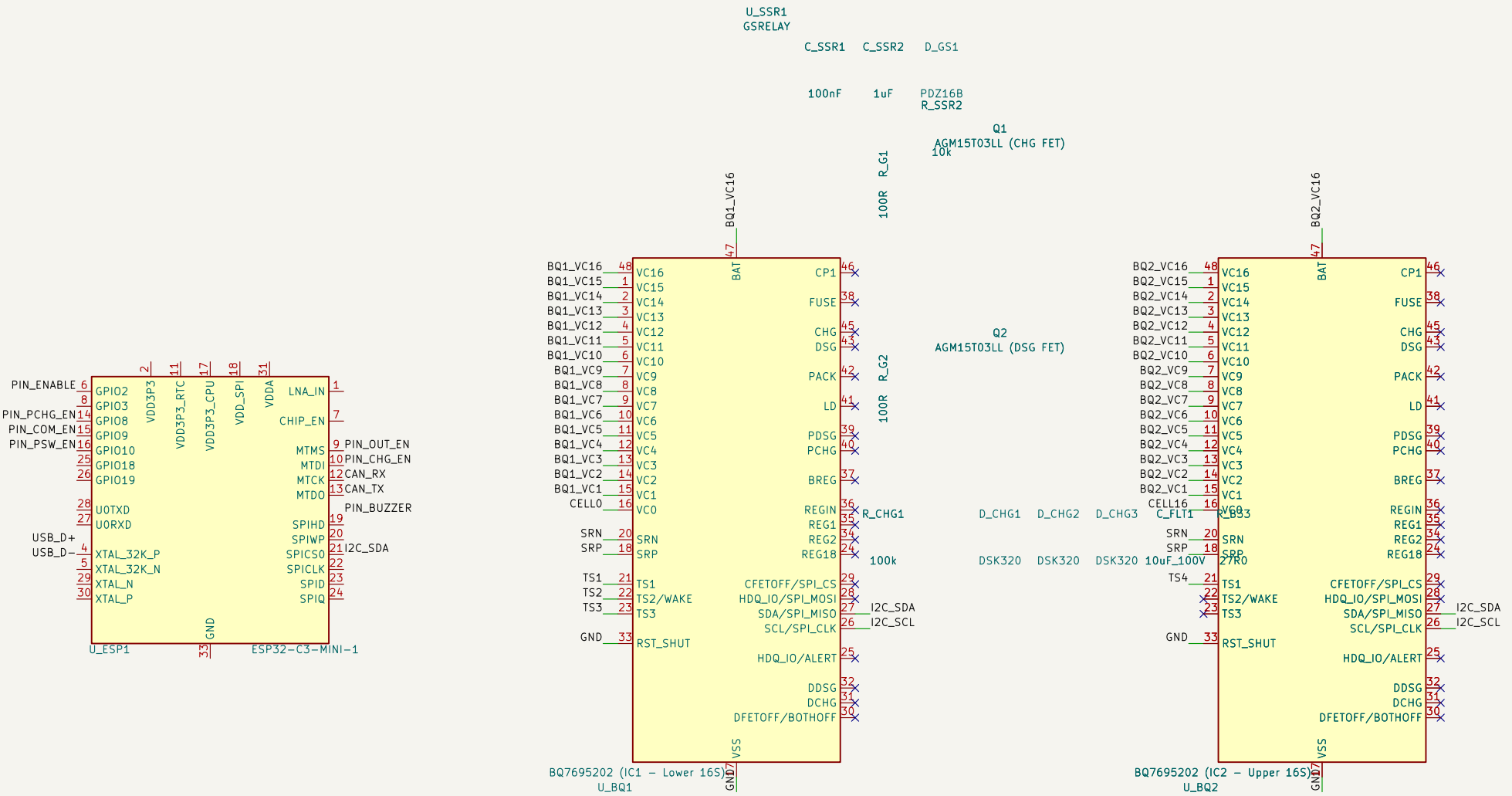
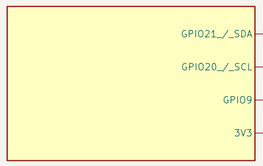

U_SSR1
GSRELAY
C_SSR1
C_SSR2
D_GS1
100nF
1uF
PDZ16B
R_SSR2
Q1
AGM15T03LL (CHG FET)
10k
R_G1
100R
Q2
AGM15T03LL (DSG FET)
R_G2
100R
BQ1_VC16
BQ2_VC16
PIN_ENABLE
PIN_PCHG_EN
PIN_COM_EN
PIN_PSW_EN
USB_D+
USB_D−
PIN_OUT_EN
PIN_CHG_EN
CAN_RX
CAN_TX
PIN_BUZZER
I2C_SDA
I2C_SCL
R_CHG1
100k
D_CHG1
D_CHG2
D_CHG3
C_FLT1
R_FLT1
DSK320
DSK320
DSK320
10uF_100V
27R0
U_ESP1
ESP32-C3-MINI-1
BQ7695202 (IC1 – Lower 16S)
U_BQ1
BQ7695202 (IC2 – Upper 16S)
U_BQ2
Sheet: /
File: qfbms32.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1

32-cell monitor interface

Charge-only BMS / 100 kHz I2C / ESP32-C3 host

HOST DOMAIN / CELL0

U205
ESP32-C3 interface



SDA_HOST
SCL_HOST
HOST_3V3

External host / logical ports

U204 TMUX154E > U_ISO ISO1640

Domain boundary inside ISO1640

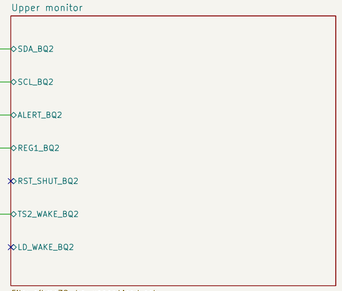
Upper I2C isolation



File: qfbms32-bq-iso-i2c.kicad_sch

UPPER DOMAIN / CELL16

Upper monitor

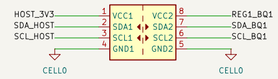


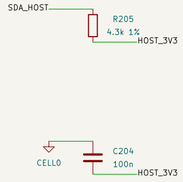
File: qfbms32-bq-upper.kicad_sch

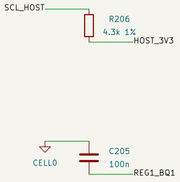
Cells 17-32 | upper address 0x08

Lower I2C / shared CELL0 return

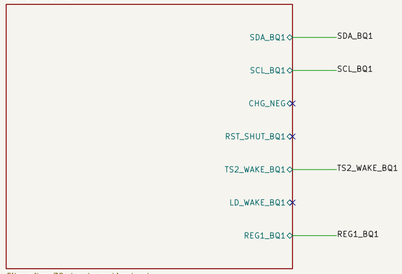
U206
ISO1640BDR







Lower monitor



File: qfbms32-bq-lower.kicad_sch

Cells 1-16 | lower address 0x08 at reset

Startup sequence

Required after lower interface closure:
1 GPIO9 high opens U204. Configure the lower address.
2 Verify lower BQ at 0x10.
3 Drive GPIO9 low to admit upper BQ at 0x08 through ISO1640.

Cold attach: verify both local REG1 rails before the first transfer.
SHUTDOWN: start host supply, press and release WAKE, wait for both BQs.

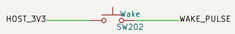
Host power / independent regulated input





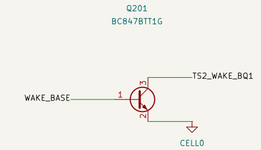
Bench source at J201. Close maintained SW201 to start the ESP even with both REG1 rails off.
Production auxiliary supply, protection and hold-up OPEN.

Manual wake / press after host power is valid



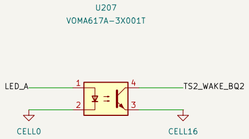






SW202 owns both wake paths. Release before requesting SHUTDOWN.
No ESP output drives the base or LED. Hot released leakage OPEN.





Schematic increment only. qfbms32.kicad_pcb is not updated from this file.

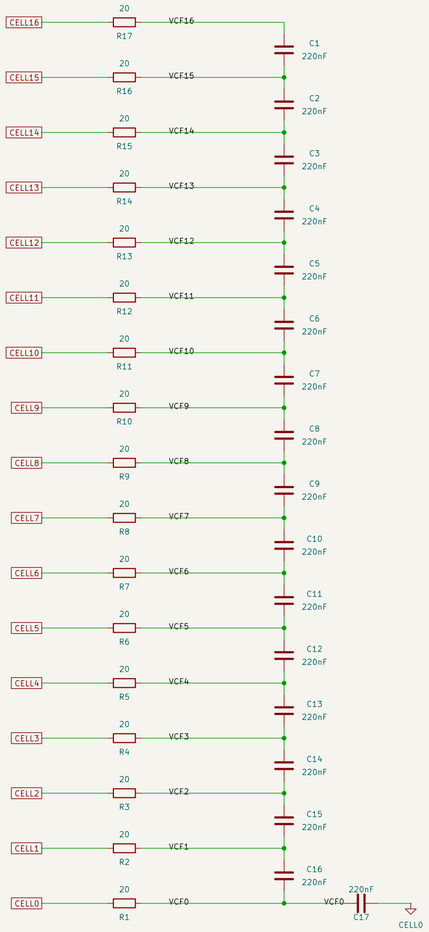
Sheet: /
File: qfbms32-bq-monitors.kicad_sch

Title: QFBMS32 BQ monitor sheets

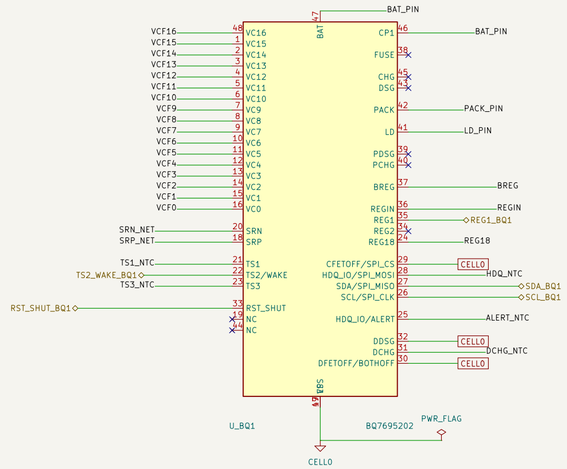
Size: A2	Date: 2026-11-04	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 1/4

Lower monitor / cells 1 to 16 / reference CELL0

01 Cell input filters

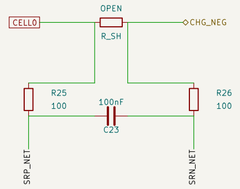


02 Monitor pinout



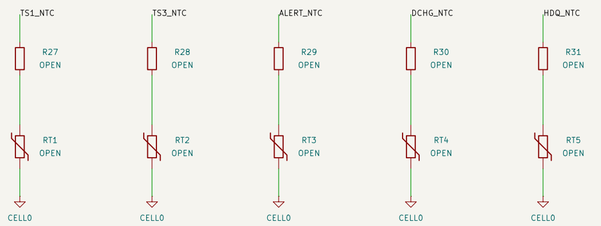
Lower SDA/SCL reach the host via U206.
Host power sequencing and bus thresholds are unresolved.

03 Current sense



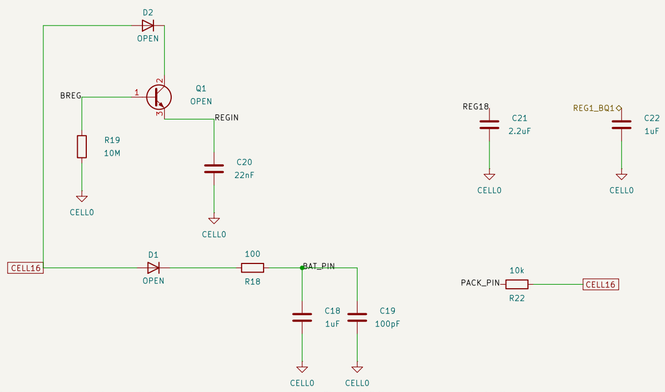
OPEN Confirm Kelvin pickup, shunt value and firmware sign/gain.

04 Temperature inputs



OPEN Firmware assumes 10 kOhm / beta 3380 and 500 ohm series correction.

05 BAT and local regulators



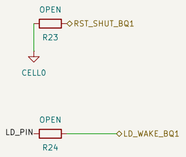
OPEN Select stack-rated diodes, REGO transistor and passive ratings. Check REGIN headroom and dissipation.

06 I2C pull-ups



OPEN Select pulls at 100 kHz. Check rail limits, sink current and rise time for this segment.

07 Wake interface



TS2 contact returns to CELL0.
Pull low, then release. REG1 can be off.
See root for driver. Hot leakage OPEN.

Charge-only monitor sheet. Do not import this netlist into qfbms32.kicad_pcb.

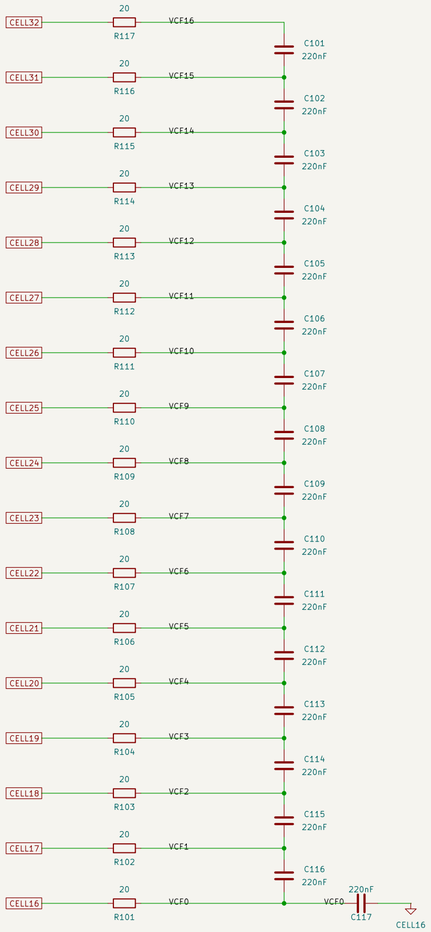
Sheet: /Lower monitor/
File: qfbms32-bq-lower.kicad_sch

Title: QFBMS32 lower monitor U_BQ1

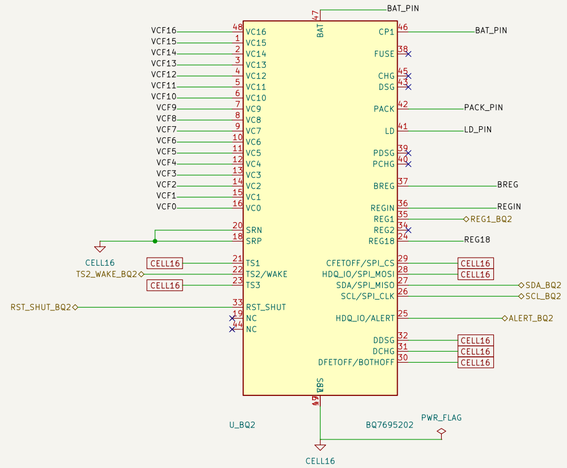
Size: A2	Date: 2026-11-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 2/4

Upper monitor / cells 17 to 32 / reference CELL16

01 Cell input filters



02 Monitor pinout



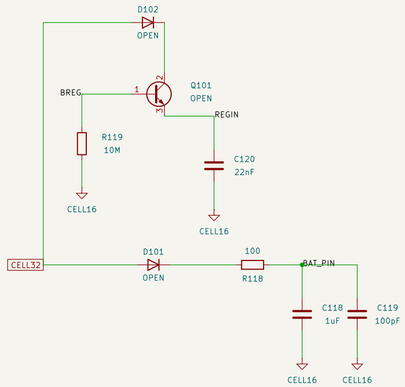
03 Current sense

SRP (18) and SRN (20) share upper VSS (17).
The current shunt is on the lower monitor.

04 Temperature inputs

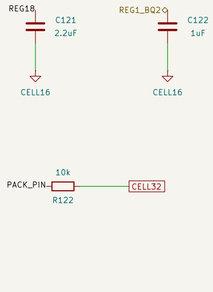
Internal temperature only. Unused sensor inputs return to CELL16.
ALERT terminates on the interface sheet.

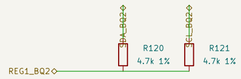
05 BAT and local regulators



OPEN Select stack-rated diodes, REGO transistor and passive ratings. Check REGIN headroom and dissipation.

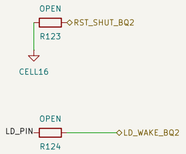
06 I2C pull-ups





OPEN Select pulls at 100 kHz. Check rail limits, sink current and rise time for this segment.

07 Wake interface



TS2 contact returns to CELL16.
Pull low, then release. REG1 can be off.
See root for driver. Hot leakage OPEN.

Charge-only monitor sheet. Do not import this netlist into qfbms32.kicad_pcb.

Sheet: /Upper monitor/
File: qfbms32-bq-upper.kicad_sch

Title: QFBMS32 upper monitor U_BQ2

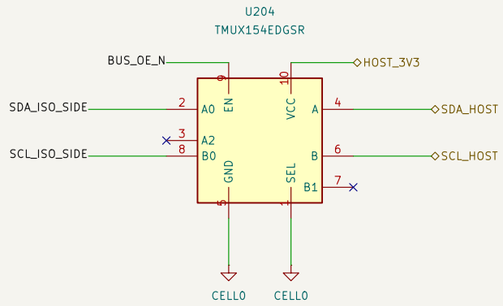
Size: A2	Date: 2026-11-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 3/4

Upper monitor communication

CELL0 host domain

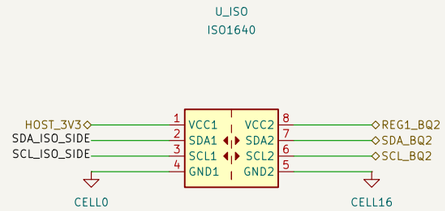
CELL16 upper domain

01 GPIO9-controlled disconnect



GPIO9 high opens both lines. Low selects A0/B0 after the lower address check.

02 ISO1640 domain boundary



OPEN Select ISO1640 orderable 8-D suffix. Check rail margins and sequencing.

03 Host pull-ups and reset bias



OPEN R201/R202: 4.7 kOhm candidate at 100 kHz. Verify rise time, sink current and 80 pF side-1 limit.
R203 biases OE high through reset. Install R204 for normal operation.



Remove R204 before strapping GPIO9 low for download. R203 then holds U204 open.

Upper ALERT stops here; host firmware does not read it.

04 Local bypass



GPIO9 is pulled high at reset. GPIO9 low selects ESP32-C3 download mode.

Sheet: /Upper I2C isolation/		
File: qfbms32-bq-iso-i2c.kicad_sch		
Title: QFBMS32 upper I2C isolation		
Size: A3	Date: 2026-11-04	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 4/4